

Answer Key — Population, Community and Interactions

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1** (b) **population** — same-kind organisms together form a population.
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- Q2** (b) **community** — all populations together make a community.
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- Q3** (b) **biotic–biotic interaction** — both frog and insects are living — biotic–biotic.
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- Q4** (b) **pollination** — this transfer is pollination.
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- Q5** (a) **Both true, R explains A** — pollination by insects lets plants form seeds — as R explains.
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- Q6** **abiotic** — oxygen/water are abiotic, so it is biotic–abiotic.
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- Q7** **community** — together they form a community.
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Short answer

- Q8** A population is a group of the same kind of organism in a habitat at a given time; a community is all the different populations sharing that habitat.
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- Q9** Biotic–biotic: a snake eats fish. Biotic–abiotic: earthworms live in moist soil (or a fish takes oxygen from water).
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- Q10** The transfer of pollen from the stamens to the carpels (by insects, birds, wind or water); it is essential for the formation of fruits and seeds.
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Long / application

- Q11** Fish eat dragonfly larvae, so a pond with fish has fewer dragonflies. Dragonflies eat bees and butterflies, so fewer dragonflies means more of these pollinators. More pollinators carry more pollen between flowers, so nearby plants produce more seeds and more flowers. This shows how biotic components are interconnected.
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- Q12** If all organisms were the same, they would all need the same food, water and space, leading to intense competition and scarcity. A variety of organisms use resources differently and depend on one another (food, pollination, decomposition), keeping the habitat balanced and sustainable.
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